Smart Laundry Solution for Dorm Students

Project Overview

1. Project concept

The Smart Laundry Solution is a connected clothing tracker and laundry management system designed for students living in dorms. It combines RFID-tagged clothing, a doorway RFID reader, and a mobile app to help students track how often they wear their clothes, decide what needs washing, schedule laundry around their classes, and keep track of clothing throughout the laundry process.

Each clothing item has its own RFID tag linked to a profile in the app. As clothing leaves the room, the system records it as worn for that day. The app uses this wear history and the student's chosen limits to organize clothes into three groups: **Good to Wear, Almost Time for Laundry, and Needs Washing**.

When laundry time arrives, students choose which clothes to wash through a swipe-based review. The system then checks that those items leave the room, provides wash and dry reminders, and verifies that they return before resetting their wear counts.

2. The problem the project addresses

Dorm students often manage laundry around classes, work, activities, and shared laundry facilities. They may lose track of which clothes they have worn, wait until too many items need washing, forget something in their room, or leave clothing behind in the laundry room.

This project brings those tasks into one system. Its goals are to:

- Reduce the effort needed to track clothing use.
- Help students plan laundry before they run out of wearable clothes.
- Fit laundry sessions into their schedules.
- Remind students when washing and drying are nearly finished.
- Detect clothing missing from an outgoing or returning laundry load.
- Prevent unrelated clothing from having its wear history reset.

3. Main system components

Component	Purpose

Individual RFID tags	Give each clothing item a unique identity.
Doorway RFID reader	Detect tagged clothing moving through the doorway.
Mobile app	Store clothing profiles, display wear counts, manage laundry loads, and send reminders.
Scheduling system	Support manually chosen laundry times or calendar-based scheduling.
Reader indicator light	Provide a visible alert, such as flashing red when a selected laundry item is missing.

The RFID reader identifies clothing; the app interprets those readings according to the current activity, such as everyday wear tracking or checking a laundry load.

4. Clothing setup and profiles

During setup, students register each clothing item and connect it to its RFID tag.

Each clothing profile includes:

- A photo of the item.
- Its color.
- Its category, such as shirt, hoodie, pants, shorts, or jeans.
- Its unique RFID identifier.
- The number of wear days allowed before washing.
- Its current wear count and laundry status.

Students can choose their own wear limit or use an app-provided default based on clothing type. For example, a student could set a shirt's limit to five wear days.

These limits are adjustable preferences. The app estimates when laundry is due using recorded wear days; it does not directly measure whether clothing is clean.

5. Everyday wear tracking

In normal operation, the system records a wear day when a registered clothing item leaves the room.

Each item can receive only one wear count per calendar day. Repeated trips through the doorway on the same day do not add more counts.

For example:

1. A student leaves the room wearing a hoodie in the morning.
2. The hoodie's wear count increases by one.
3. The student returns and leaves again later.
4. The hoodie receives no additional count that day.
5. If it leaves the room the following day, it can receive another count.

This keeps the tracker focused on days worn rather than the number of doorway crossings.

The initial concept assumes that clothing leaving the room during normal tracking has been worn. Handling clothes carried out for other reasons is a behavior to refine during development.

6. Clothing status and visual tracking

The app displays each item's progress toward its wear limit using a green, yellow, and red timeline or progress indicator.

Status	Meaning
Green — Good to Wear	The item remains comfortably below its wear limit.
Yellow — Almost Time for Laundry	The item is approaching its wear limit.
Red — Needs Washing	The item has reached or exceeded its wear limit.

The clothing tracker has three matching tabs so students can quickly browse each group. Every item displays its photo, current wear count, and limit—for example, **"4 of 5 wear days."**

The exact point at which an item becomes yellow will be defined during development and should work sensibly for different wear limits.

7. Laundry scheduling

After clothing setup, the app prompts students to connect Google Calendar for scheduling support. Students can choose between two scheduling approaches.

Manual scheduling: Students select their own laundry days and times, such as every Sunday afternoon or a specific upcoming evening.

Automatic scheduling: With calendar access, the app checks for available time within the student's selected laundry hours. When enough clothing is close to or beyond its wear limit, it identifies a suitable opening and creates a laundry session.

Automatic scheduling should consider:

- How many items need washing or are approaching their limits.
- The student's calendar commitments.

- The hours during which laundry can be done.
- The expected time needed for washing and drying.
- Any existing laundry sessions.

The number of items that triggers scheduling, the session duration, and how schedule changes are handled remain adjustable project rules.

8. App home page

The home page serves as the main dashboard. It provides access to:

- **Upcoming laundry sessions:** View planned laundry days and times.
- **Clothing tracker:** Browse clothing, wear counts, and status.
- **Ready for Laundry:** Start reviewing clothes for the next load.
- **Active laundry session:** Follow the current load through washing, drying, and return.
- **Alerts:** See missing-item notices and timer reminders.

The dashboard changes with the laundry session so students can easily find the next action.

9. “Ready for Laundry” swipe feature

Before a laundry session, the student taps **Ready for Laundry** to review clothing through a Tinder-style interface.

Clothes appear in priority order:

1. Red items that need washing.
2. Yellow items approaching their limits.
3. Green items still available to wear.

Each card shows the clothing photo, current wear count, wear limit, and status.

- **Swipe right — Wash it:** Add the item to the laundry cart.
- **Swipe left — Skip it:** Leave the item out of the load and move its displayed wear count back by one day.

Under the initial concept, skipping a five-day item would reduce its count to four days. This gives students a way to postpone washing based on their own judgment. Counts should never fall below zero.

Because skipping does not physically clean an item, this count-reduction rule is a product decision to revisit as the project develops.

10. Laundry cart and departure check

After reviewing the clothing, the student taps **Done**. The app shows a summary of all items selected for washing.

Once the load is confirmed, the reader enters **laundry departure mode**. In this mode, only items in the confirmed laundry cart affect the load check.

As selected items leave the room:

- The system marks them as detected.
- Their status changes to **Being Washed**.
- The app tracks which selected items are still missing.

If an item remains unscanned after the departure check, the app sends a forgotten-item notification and the reader flashes red. The notification should identify the missing item using its photo and profile.

The system needs a short scan window or an explicit completion check before declaring an item missing, since a full load may take several moments to pass through the doorway.

11. Washing and drying tracking

Once all selected clothing has left the room, the home page displays a **Track Your Laundry** prompt.

After loading and starting the washer, the student:

1. Taps **Start Washing**.
2. Enters the wash-cycle duration.
3. Starts the app timer.

The app sends a notification five minutes before the cycle ends and another when the timer finishes.

After transferring the load to the dryer, the student taps **Start Drying**, enters the drying duration, and starts the next timer. The app again sends notifications five minutes before completion and at completion.

In this version, cycle tracking depends on student-entered timers. Direct communication with washing machines and dryers is outside the initial concept.

12. Return check and wear-count reset

After drying finishes, the system enters **laundry return mode** and looks for the clothing associated with that laundry session.

As each expected item returns through the doorway:

- The system marks it as returned.
- Its wear count resets to zero.
- Its status returns to **Good to Wear**.

If some items return and others remain missing after the return check, the app sends a notification identifying the missing clothing.

Items that have not returned remain unresolved in the laundry session rather than being automatically marked clean.

13. Protecting unrelated clothing

A central requirement is that laundry scans affect **only the items assigned to the active laundry cart**.

For example, a student may wear a hoodie while carrying a laundry basket. If that hoodie is not in the confirmed load, detecting its tag must not reset its wear count or mark it as washed.

The same protection applies when the student returns with clean laundry.

Technically, the reader may still detect nearby tags. The app must filter those readings so only the expected laundry items update the active session. Everyday wear counting should also remain separate from laundry verification to prevent a laundry departure from adding an accidental wear day.

14. Main operating states

State	System behavior
Normal tracking	Record at most one wear day per item per day.
Laundry selection	Let the student review clothing and build a cart.
Departure check	Verify that selected items leave the room.
Washing	Track the student-started wash timer and reminders.
Drying	Track the student-started drying timer and reminders.
Return check	Verify returning items and reset their wear counts.
Completed or unresolved	Close the session when all items return, or show missing items.

15. Hardware and data flow

The system uses a **YRM1003 RFID reader** to scan the RFID tags attached to clothing. The reader passes the detected tag information to an **ESP32**, which stores the scan data locally and uses Wi-Fi to send it to the cloud.

The mobile app retrieves this data from the cloud to update clothing wear counts, track laundry progress, and display missing-item alerts.

Data flow: Clothing RFID tag → YRM1003 RFID reader → ESP32 → Wi-Fi → Cloud → Mobile app.